

Education

PhD in Applied Physics, Northwestern University 2017 - Present

MSci in Physics, University College London (UCL) 2012 - 2016

- First Class Honours

Research

Doctoral Researcher, Northwestern University 2017 - Present

- Discovering and evaluating new materials for renewable energy applications
- Developing machine learning models to predict materials properties
- Collaborating with experimental research groups and industry researchers
- Managing organization of group meetings, software licensing and distribution, allocation of computational resources, providing initial resources to new students
- Managing the writing of yearly proposals for the award of computational resources by the US DOE

Intern, European Council for Nuclear Research (CERN) 2015 - 2016

- Characterizing new-generation particle detectors on both a theoretical and experimental level

Masters Researcher, University College London (UCL) 2015 - 2016

- Modeling the quantum mechanical processes involved in nuclear magnetic resonance
- Developing piezoelectric-based energy harvesting devices

Intern, National Institute of Nuclear Physics (INFN) 2013

- Performing data-analysis and performance optimization of particle detectors

Teaching

Learning Community Discussion Leader, Northwestern University 2021 - 2022

- Developing curriculum, organizing and leading discussion sessions themed around evidence based undergraduate STEM teaching

Instructor, Pueblo Science 2021 - 2022

- Demonstrating hands-on STEM lesson plans for secondary school teachers

Coding Club Mentor, Boys and Girls Club Chicago 2019 - 2020

- Developing course materials and providing individual mentorship to high-school students

Teaching Assistant, Northwestern University 2019

- Leading undergraduate students through basic theory, experiments and data analysis in introductory physics, assessing weekly written reports and providing individualized feedback and guidance

Service and Outreach

Applied Physics Mentor, Northwestern University 2021 - 2022

- Advising first year students in the applied physics graduate program

Undergraduate Research Mentor, Northwestern University 2020 - 2021

- Teaching skills, discussing ideas, setting goals, providing feedback

Accountant, Mosaic Coop 2019 - 2020

- Pro-bono accounting for a non-profit housing organization

Professional Development

Northwestern Center for Integration of Research Teaching and Learning Associate Certificate	2022
Kellogg Management for Scientists and Engineers Certificate	2022
Northwestern Integrated Data Science Certificate	2022

Publications and Conferences

Publications

- Shen et al “*Reflections on One Million Structures in the Open Quantum Materials Database (OQMD)*” J. Phys. Mater. (2022)
- Qian et al “*Remarkable Properties and Performance of $\text{CaTi}_{0.5}\text{Mn}_{0.5}\text{O}_{3.8}$ for Solar-Driven Thermochemical Hydrogen Production*” Matter (2021)
- Qian et al “*Favorable Redox Thermodynamics of $\text{SrTi}_{0.5}\text{Mn}_{0.5}\text{O}_{3.8}$ in Thermochemical Water Splitting*” Chemistry of Materials (2020)
- Baldassarri et al “*Signal Formation in Irradiated Silicon Detectors*” Nuclear Instruments and Methods in Physics Research Section A (2017)
- Cartiglia et al “*Tracking in 4 Dimensions*” Nuclear Instruments and Methods in Physics Research Section A (2017)
- Sadrosinzi et al “*Ultra Fast Silicon Detectors*” Nuclear Instruments and Methods in Physics Research Section A (2016)

Conferences

- 2022 APS March Meeting
- 2021 APS March Meeting
- 2020 APS March Meeting
- 2016 Vienna Conference on Instrumentation

Manuscripts

- Baldassarri et al “*Accuracy of DFT Computed Oxygen Vacancy Formation Energies and High-Throughput DFT Search of Solar Thermochemical Water Splitting Compounds*” (under review at Phys Rev Mat)
- Baldassarri et al “*Oxygen Vacancy Formation Energy in Metal Oxides: High Throughput Studies and Machine Learning Predictions*”
- Baldassarri et al “*DFT Study of Vibrational Entropies of Vacancy Formation in Metal Oxides*”
- He et al “*Influence of Exchange-Correlation Functional on Oxygen Vacancy Formation Energy of Transition Metal Oxide*”
- He et al “*Stability Map of Double Perovskite Oxides*”
- He et al “*Stability Map of Pyrochlore Oxides*”
- Liu et al “*High-throughput screening of short- and long-range order and electrochemical performance for rocksalt-type Li-ion battery cathode materials design*”

Software and Programming

- Current work: Python, VASP
- Previous experience: Stata, Gephi, Java, C++, Matlab, IDL, Labview, Autocad

Languages

- English - Proficient , Italian - Proficient , French - Conversational , Spanish - Conversational